SAP Procure-to-Pay — Capstone Project

ABC Industrial Supplies Ltd.
Quarterly Production Materials Supply Order

COMPANY	ABC Industrial Supplies Ltd.
INDUSTRIES	Quarterly supply order for production materials
SAP MODULE	SAP OTC / Order-to-Cash (O2C) Cycle
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BATCH / PROGRAM	2023,EEE / C_TS462 - SAP Certified Associate - SAP S/4HANA Cloud Private Edition, Sales
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COURSE	SAP Order to Cash — Capstone Project
SOURCES	Fictitious company master data, SAP S/4HANA training environment logs, and standard SAP functional configuration guides.

Problem Statement: Inefficient Manual Procurement

Current manual and spreadsheet-based processes cause:

- **Visibility Gaps:** No real-time tracking, leading to stockouts or duplicate orders.
- **Data Mismatches:** Inconsistent entries across Finance and Warehouse departments.
- **Bottlenecks:** Delayed physical approvals slowing down raw material procurement.
- **Compliance Risks:** Lack of audit trails for financial and vendor spending history.

Solution: SAP Procure-to-Pay (P2P)

This project implements the **SAP Materials Management (MM)** framework to automate the lifecycle from requisition to payment. By using a single digital workflow, it ensures data integrity and establishes a "single source of truth" across all departments.

Key Features

- **Automated Requisitions:** Streamlines internal requests to prevent duplicate spending.
- **Centralized Vendor Data:** Standardizes payment terms (Net 30) and Incoterms (FOB).
- **Integrated Inventory:** Instant stock updates via Goods Receipt (MIGO).
- **Three-Way Match:** Auto-verifies PO, Receipt, and Invoice for financial accuracy.
- **Org Mapping:** Links Company Codes, Plants, and Purchasing Orgs for unified reporting.
- **Audit Trail:** Maintains a complete digital history for transparent compliance.

FIG 1: Header Data Configuration

Description: This figure illustrates the **Header Data** for a new Purchase Order (Transaction ME21N).

Key Fields: It defines the **Vendor (ABC Industrial Supplies Ltd.)**, the **Company Code (1000)**, and **Purchasing Organization (1000)**.

Function: This section establishes the legal and organizational framework for the entire procurement transaction, including payment terms (Net 30 Days) and Incoterms (FOB).

FIG 1: Header Data Configuration

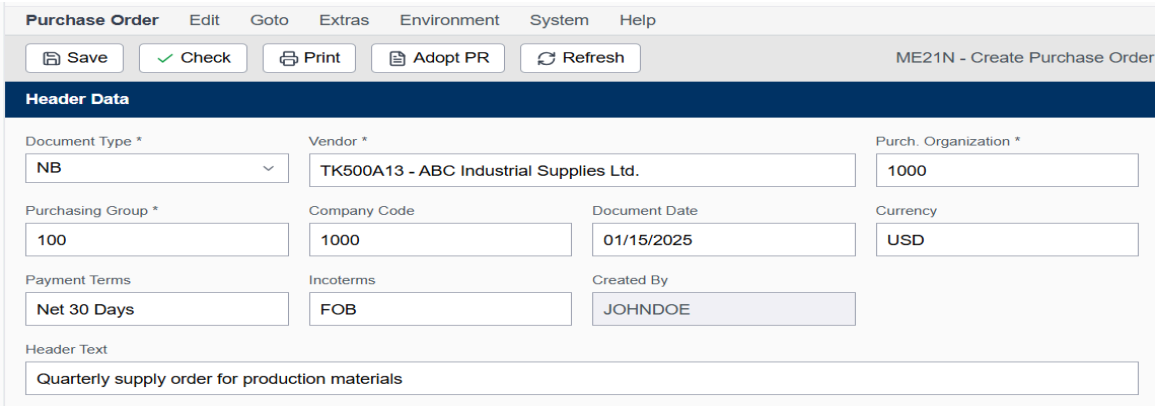


FIG 2: Item Overview and Material Selection

Item Overview Items: 3

☐	Item	Material	Short Text	PO Quantity	Unit	Delivery Date	Net Price	Currency	Plant	Storage Loc.
☐	10	100-100	Steel Pipes DN 50	100	EA	02/15/2025	50.00	USD	TH000	0001
☐	20	M-01	Hydraulic Pump Assembly	10	PC	02/20/2025	500.00	USD	TH000	0002
☐	30	RAW-300	Industrial Lubricant Oil	500	LTR	02/10/2025	12.50	USD	TH000	0003

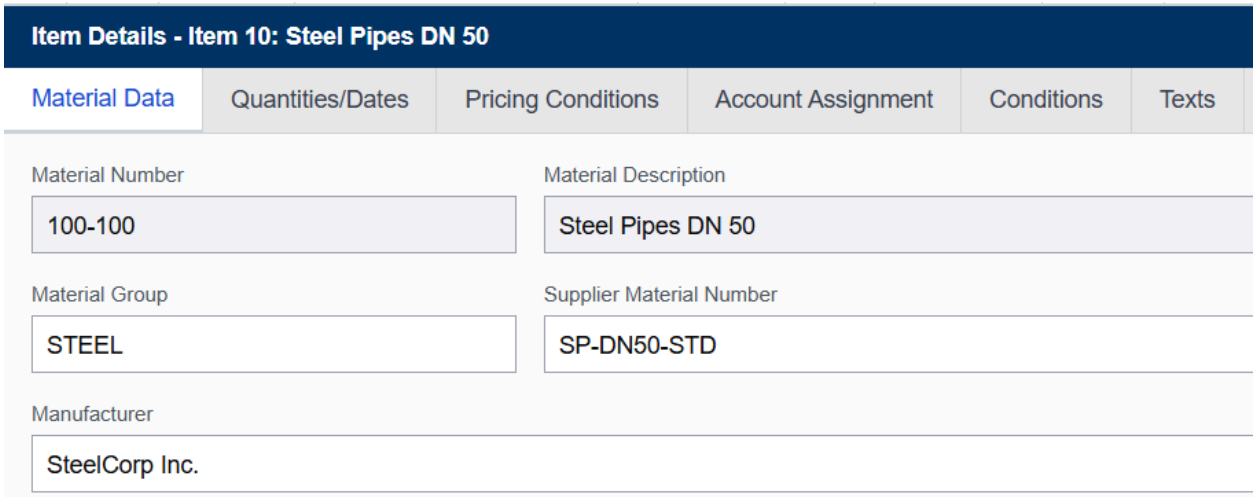
Description: The **Item Overview** provides a summary of the materials being procured.

Key Fields: It lists multiple line items, including **Steel Pipes (Item 10)**, **Hydraulic Pumps (Item 20)**, and **Lubricant Oil (Item 30)**.

Function: It captures the specific **PO Quantity**, **Net Price**, and the receiving **Plant (TH000)** and **Storage Location (0001)** for each material, allowing for bulk procurement in a single document.

FIG 3: Granular Item Details

- **Description:** This figure shows the **Item Details** specific to Item 10 (Steel Pipes).
- **Key Fields:** The **Material Data** tab displays the **Material Group (STEEL)** and the **Manufacturer (SteelCorp Inc.)**.
- **Function:** This level of detail ensures the warehouse and vendor are aligned on the specific technical requirements and manufacturer of the material, reducing errors during the Goods Receipt stage.



Tech Stack

TECHNOLOGY	Purpose in P2P Cycle
SAP S/4HANA / ERP	The core Enterprise Resource Planning system used to host and integrate all procurement business processes.
SAP MM (Materials Management)	The functional module used managing inventory, vendor master data, and purchasing lifecycle
SAP FI (Financial Accounting)	Facilitates the integration of procurement with the General Ledger (G/L) and Cost Centers for accurate financial reporting.
Fiori / GUI Interface	execute transactions like ME21N and MIGO (Goods Receipt).

Item Details - Item 10: Steel Pipes DN 50

Material Data | Quantities/Dates | Pricing Conditions | Account Assignment | Conditions

PO Quantity	Order Unit	Delivery Date
100	EA	02/15/2025
Plant	**Storage Location**	**GR Processing Time**
TH000	0001	3 Days

Item Details - Item 10: Steel Pipes DN 50

Material Data | Quantities/Dates | Pricing Conditions | Account Assignment | Conditions

G/L Account	Cost Center	Internal Order
400100	CC-1000	-
WBS Element	**Asset Number**	
-	-	

Item Details - Item 10: Steel Pipes DN 50

Material Data | Quantities/Dates | Pricing Conditions | Account Assignment | Conditions

Net Price	Price Unit	Currency
50.00	1	USD
Gross Price	**Discount**	**Effective Price**
52.00	3.85%	**50.00**
Tax Code	**Tax Amount**	**Total Item Value**
V1	5.00	**5000.00 USD**

In the system logistics and financial integration (Figures 4, 5, and 6), the solution provides granular control over the supply chain:

- **Logistics & Scheduling (Fig 4):** Precisely manages **PO Quantities (100 EA)** and **Delivery Dates (02/15/2025)**, ensuring materials arrive at **Plant TH000** exactly when needed.
- **Pricing Precision (Fig 5):** Automates complex calculations, applying a **3.85% discount** to the **Gross Price ($52.00)** to achieve an **Effective Price of $50.00**, totaling a controlled value of **$5000.00 USD**.
- **Financial Integration (Fig 6):** Directly maps procurement costs to **G/L Account 400100** and **Cost Center CC-1000**, ensuring that every purchase is automatically reflected in the company's financial records without manual entry.

Unique Points

- **Cross-Functional Integration:** The project demonstrates a seamless link between logistics (MM) and finance (FI) by mapping specific G/L accounts and Cost Centers directly within the Purchase Order.
- **Dynamic Pricing Architecture:** Utilizes automated pricing conditions that calculate effective costs by applying tiered discounts and tax codes (V1) to gross prices in real-time.
- **Granular Data Integrity:** By configuring specific Supplier Material Numbers and Manufacturer details, the solution minimizes the risk of procuring incorrect parts for technical operations.
- **Operational Scheduling:** Features a defined "GR Processing Time" of 3 days, allowing the organization to plan production schedules based on realistic material availability dates.

Future Improvements

Mobile Approvals: Fiori-based apps for on-the-go PR/PO authorization.
Ariba Integration: Digital supplier portal to automate invoice entry.
AI Analytics: S/4HANA predictive stock replenishment and shortage alerts.
ESG Tracking: Green procurement fields for carbon footprint monitoring.